

Assignment 1: Linear Regression and Classification

IE203 Data Mining

This assignment is entirely practical: every question is answered with code and with short comments/interpretation of your results (written in a markdown cell). No written derivations or proofs are required anywhere in this assignment.

PART A: Linear Regression

A companion Jupyter notebook (`A1.LinearRegression.Template.ipynb`) contains the same problems with empty code cells for you to fill in. Submit your completed notebook.

Datasets needed: `Advertising.csv`, `fire_incidents.csv` (provided together with this notebook).

P1) Multiple Linear Regression — the Advertising dataset

A company wants to understand how its advertising budget on three channels (**TV**, **Radio**, **Newspaper**, all in thousands of dollars) relates to **Sales** (in thousands of units) in 200 different markets. The data is in `Advertising.csv`.

- P1.1) **Load and explore.** Load the dataset (drop the row-index column if there is one). Print the first 5 rows, the shape, and the summary statistics (`.describe()`). Then produce a scatterplot of **Sales** against each of the three predictors, and the correlation matrix of all four variables. Comment briefly on what these suggest about which channel(s) might matter most, and whether any pair of predictors looks correlated with each other.
- P1.2) **Simple linear regressions.** Fit three *separate* simple linear regressions: $\text{Sales} \sim \text{TV}$, $\text{Sales} \sim \text{Radio}$, $\text{Sales} \sim \text{Newspaper}$. For each, report the estimated coefficient, its p-value, and the model's R^2 .
- P1.3) **Multiple linear regression.** Fit a single multiple regression $\text{Sales} \sim \text{TV} + \text{Radio} + \text{Newspaper}$. Report the coefficients, p-values, overall R^2 , and residual standard error (RSE). Compare the coefficient and p-value of **Newspaper** here to what you found in P1.2 — what changed, and why?
- P1.4) **A reduced model.** Refit the model dropping **Newspaper**. Compare its R^2 and RSE to the full model. Is dropping **Newspaper** a good idea?

P2) Generate your own dataset: small simulation study

Use the default random number generator with **seed set to 7** throughout this problem, so your results are reproducible.

- P2.1) **Generate X.** Draw $n = 120$ observations $X \sim N(0, 1.5^2)$.
- P2.2) **Generate the noise.** Draw $\varepsilon \sim N(0, 0.36)$ (mean 0, variance 0.36, i.e. standard deviation 0.6), 120 observations.
- P2.3) **Generate y.** Using X and ε , generate

$$y = 1.5 + 0.7X + \varepsilon.$$

- P2.4) **Scatterplot.** Plot y against X .
- P2.5) **Fit a simple linear regression.** Fit $y \sim X$. Comment on the model obtained — how do $\hat{\beta}_0, \hat{\beta}_1$ compare to the true $\beta_0 = 1.5, \beta_1 = 0.7$?
- P2.6) **Is a quadratic term needed?** Fit $y \sim X + X^2$. Is there evidence that the quadratic term improves the fit? Explain.
- P2.7) **Monte Carlo simulation.** Repeat the data-generating process (P2.1–P2.3) and the fit (P2.5) **1000 times** ($n = 120$ each time), storing $\hat{\beta}_1$ each time. Plot a histogram of the 1000 estimates; report their mean and standard deviation; compare the mean to the true $\beta_1 = 0.7$.
- P2.8) **Effect of sample size.** Repeat the Monte Carlo experiment for $n = 20, 120, 1000$. Report mean/std of $\hat{\beta}_1$ for each; plot the three histograms together. How does the *spread* change with n ? Does the *mean* change the same way?
- P2.9) **Effect of noise level.** Fix $n = 120$; repeat for $\sigma^2 = 0.09$ and $\sigma^2 = 2.25$, and compare to your $\sigma^2 = 0.36$ result from P2.7. What role does the noise level play?

P3) Confounded predictors: fire damage

A city fire department wants to know whether sending **more firefighters** to a fire causes **more property damage**. Records of 400 incidents are in `fire_incidents.csv`: `FireSize_sqm`, `NumFirefighters`, `PropertyDamage_k`.

- P3.1) **Naive single-predictor regression.** Draw a scatterplot and fit a simple regression of `PropertyDamage_k` on `NumFirefighters`. Report coefficient, p-value, R^2 . What would you conclude?
- P3.2) **A second single-predictor regression.** Repeat 3.1, but with `FireSize_sqm` as predictor.
- P3.3) **Are the two predictors related?** Compute the correlation between `NumFirefighters` and `FireSize_sqm` (and the full correlation matrix among all the predictors in the dataset). What do you find, and why might a dispatch protocol produce this pattern?
- P3.4) **Multiple regression: controlling for fire size.** Fit a multiple regression `PropertyDamage_k ~ NumFirefighters + FireSize_sqm`. How does the coefficient/significance of `NumFirefighters` compare to P3.1?
- P3.5) **Explain the discrepancy.** In up to 4–6 sentences, explain why the P3.1 conclusion is misleading, using the concept of a *confounding variable*. What would you tell the fire department if asked “should we send fewer firefighters to reduce damage?”

PART B: Classification

A companion Jupyter notebook (`A2_Classification_Template.ipynb`) contains the same problems with empty code cells for you to fill in. Submit your completed notebook.

Data needed: `drug.csv` (provided with this notebook). Problems 2 and 3 use datasets available directly through `scikit-learn`, so no extra files are needed.

P1) Multi-class classification — which drug will a patient respond to?

`drug.csv` has 200 patients, each best-treated by one of five drugs (`drugA`, `drugB`, `drugC`, `drugX`, `drugY`), which is the target response. The predictors are `Age`, `Sex`, `BP` (LOW/NORMAL/HIGH), `Cholesterol` (NORMAL/HIGH), and `Na_to_K`.

- P1.1) **Load and explore.** Load the dataset and examine the relationship between `Na_to_K` and `Drug`. Include an appropriate visualization and briefly comment on any patterns you observe.
- P1.2) **Preprocess the data.** Prepare the categorical predictors appropriately for classification. Briefly describe the preprocessing choices you make.
- P1.3) **LDA and QDA.** Fit Linear Discriminant Analysis and Quadratic Discriminant Analysis models. Evaluate both models using accuracy and confusion matrices, and compare their performance.
- P1.4) **Logistic regression and Naive Bayes.** Fit multinomial logistic regression and Gaussian Naive Bayes classifiers. Evaluate them using the same performance measures as in P1.3. Comment briefly on any differences you observe.
- P1.5) **KNN and the choice of K.** Investigate K-Nearest Neighbors classifiers for $K = 1, \dots, 30$. Account appropriately for differences in predictor scales¹. Examine how accuracy changes with K , identify the best-performing value, and explain why feature scaling is particularly important for KNN.
- P1.6) **Compare all five models.** Compare LDA, QDA, logistic regression, Naive Bayes, and the selected KNN model using accuracy and weighted F1. Summarize the results in a table and discuss what the differences in performance suggest about the structure of the classification problem.

P2) Binary classification, feature significance, and class imbalance — breast tumor diagnosis

Use `sklearn.datasets.load_breast_cancer`, restricted to the 10 mean predictors:

`mean radius`, `mean texture`, `mean perimeter`, `mean area`, `mean smoothness`, `mean compactness`, `mean concavity`, `mean concave points`, `mean symmetry`, `mean fractal dimension`.

Target: malignant (0) / benign (1).

¹Note that the predictor values must be normalized to the same range to use KNN properly. This aspect was not discussed in class.

- P2.1) **Load and prepare the data.** Construct the predictor matrix and target using the 10 mean measurements. Prepare the predictors appropriately for the classification methods that follow.
- P2.2) **Logistic regression coefficients and significance.** Fit a logistic regression model using the 10 predictors and examine the estimated coefficients and their statistical significance. Identify the 2–3 predictors showing the strongest evidence of association with the response.
- P2.3) **Backward elimination.** Construct a reduced model by removing the three predictors showing the weakest statistical evidence in P2.2. Compare the reduced model with the full model. Did the apparent significance of any remaining predictors change noticeably? Explain why this might occur.
- P2.4) **Compare LDA, QDA, Naive Bayes, logistic regression, and KNN.** Fit all five classifiers using the full 10-predictor dataset. For KNN, investigate $K = 1, \dots, 25$ and select a suitable value based on performance. Compare the models using accuracy, precision, recall, and F1, treating malignant cases (class 0) as the positive class.
- P2.5) **Class imbalance: is accuracy trustworthy here?** Compare the classification models with a simple classifier that always predicts benign. Discuss whether accuracy alone gives a sufficient picture of model performance.

Now consider a setting in which only about 10% of observations are malignant. How would this affect the usefulness of accuracy as the main evaluation metric? Which performance measure(s) would become especially important if missing a malignant case is substantially more costly than incorrectly flagging a benign case?

P3) KNN in depth — decision boundaries and choosing K

A synthetic, purely 2-D dataset will be used to examine how KNN behavior depends on K .

- P3.1) **Generate and explore the data.** Generate 400 observations using `sklearn.datasets.make_moons` with `noise=0.25` and `random_state=7`. Prepare the predictors appropriately and visualize the observations by class.
- P3.2) **Decision boundaries for different K .** For $K \in \{1, 5, 15, 50\}$, visualize the decision regions produced by KNN together with the observations. Describe how the shape and complexity of the decision boundary changes as K increases.
- P3.3) **Effect of K on classification error.** Investigate KNN models for $K = 1, \dots, 50$. Examine how classification error changes as K varies, identify the value of K with the lowest error, and relate this result to the decision boundaries observed in P3.2.
- P3.4) **KNN vs. parametric classifiers.** Fit LDA, QDA, and logistic regression on the same dataset and compare their performance with the selected KNN model. Explain the differences you observe in relation to the shape of the data and the assumptions made by the different classifiers.